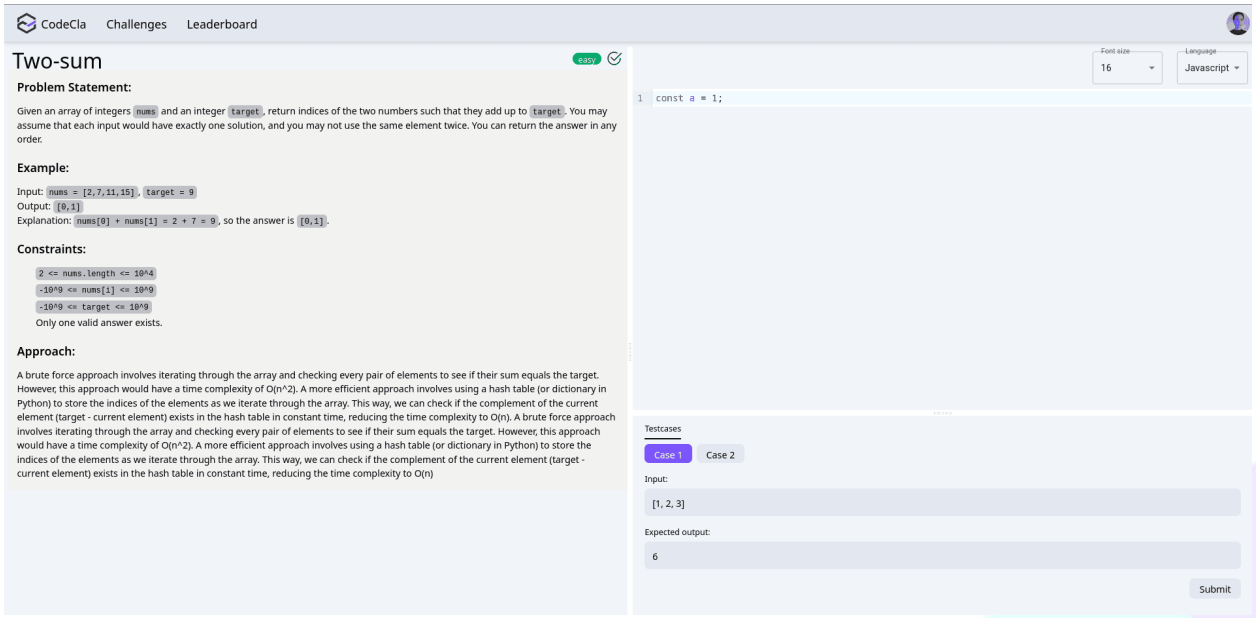
Now, let's dive into the fun part! You'll be implementing the workspace where the coder can:

- Read the challenge description
- Write the code for that challenge
- Inspect the test cases
- Submit their solution to be graded.



Tasks

1. Implementing Components

1.1. Workspace Component

Implement a Workspace page component that contains all the other components.

- The component splits the screen into two halves, the first half for a component called ChallengeDescription and the second for a component called Playground that we are going to detail in a moment.
- To split the screen, you can make use of Split component provided by react-split. It has a drag feature, so that you can play with size of each half.

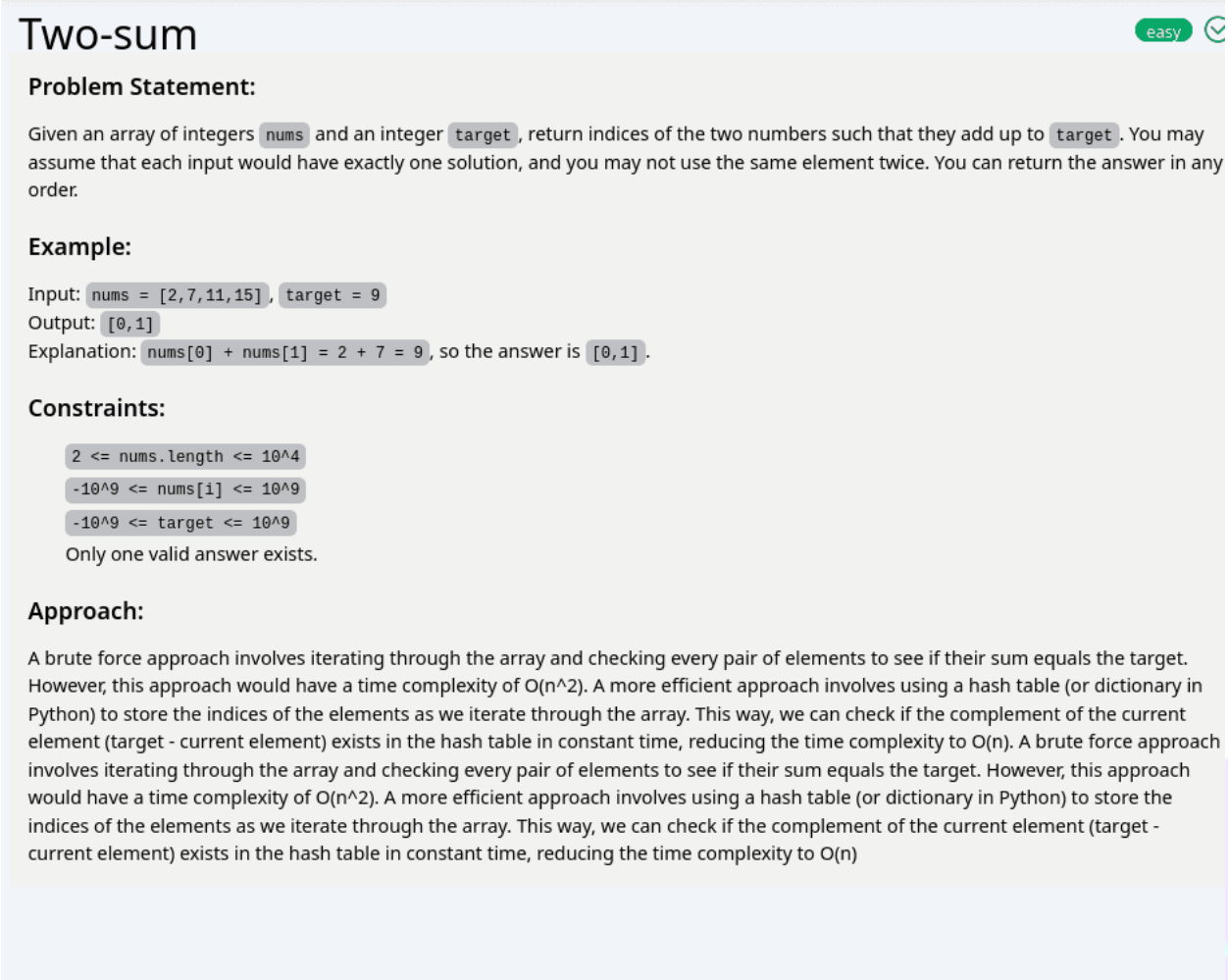
1.2. ChallengeDescription Component

- The first half of the screen is for the ChallengeDescription component that renders a markdown formatted text. So it should allow us to preview that markdown.
- To render makrdown content, you can make use of this react-markdown-preview package. It's easy and simple to use.
- For the data, you can start with this data, but as we move forward you are going to implement the backend and integrate it with this UI with real data.

```
let challenges = [
  {
    id: 123,
    title: "Two-sum",
    description: `
### Problem Statement:
Given an array of integers \`nums\` and an integer \`target\`, return
indices of the two numbers such that they add up to \`target\`. You
may assume that each input would have exactly one solution, and you
may not use the same element twice. You can return the answer in any
order.
### Example:
Input: \`nums = [2,7,11,15]\`, \`target = 9\`
Output: \`[0,1]\`
Explanation: \`nums[0] + nums[1] = 2 + 7 = 9\`, so the answer is
\`[0,1]\`.
### Constraints:
- \`2 <= nums.length <= 10^4\`
- \`-10^9 <= nums[i] <= 10^9\`
- \`-10^9 <= target <= 10^9\`
- Only one valid answer exists.
### Approach:
A brute force approach involves iterating through the array and
checking every pair of elements to see if their sum equals the
target. However, this approach would have a time complexity of
```

```
O(n^2). A more efficient approach involves using a hash table (or
dictionary in Python) to store the indices of the elements as we
iterate through the array. This way, we can check if the complement
of the current element (target - current element) exists in the hash
table in constant time, reducing the time complexity to O(n).
`, // md content,

    difficulty: "Easy",
    category: "arrays",
    status: "Completed",
    tests: [
      {
        id: 1,
        input: {}, // Not used here
        inputText: `[1, 2, 3]`,
        output: {}, // Not used here
        outputText: `6`,
      },
      {
        id: "test_2",
        input: {}, // Not used here
        inputText: `[2, 3]`,
        output: {}, // Not used here
        outputText: `5`,
      },
    ],
  },
];
```



Two-sum

easy

Problem Statement:

Given an array of integers `nums` and an integer `target`, return indices of the two numbers such that they add up to `target`. You may assume that each input would have exactly one solution, and you may not use the same element twice. You can return the answer in any order.

Example:

Input: `nums = [2,7,11,15]`, `target = 9`
Output: `[0,1]`
Explanation: `nums[0] + nums[1] = 2 + 7 = 9`, so the answer is `[0,1]`.

Constraints:

- `2 <= nums.length <= 10^4`
- `-10^9 <= nums[i] <= 10^9`
- `-10^9 <= target <= 10^9`
- Only one valid answer exists.

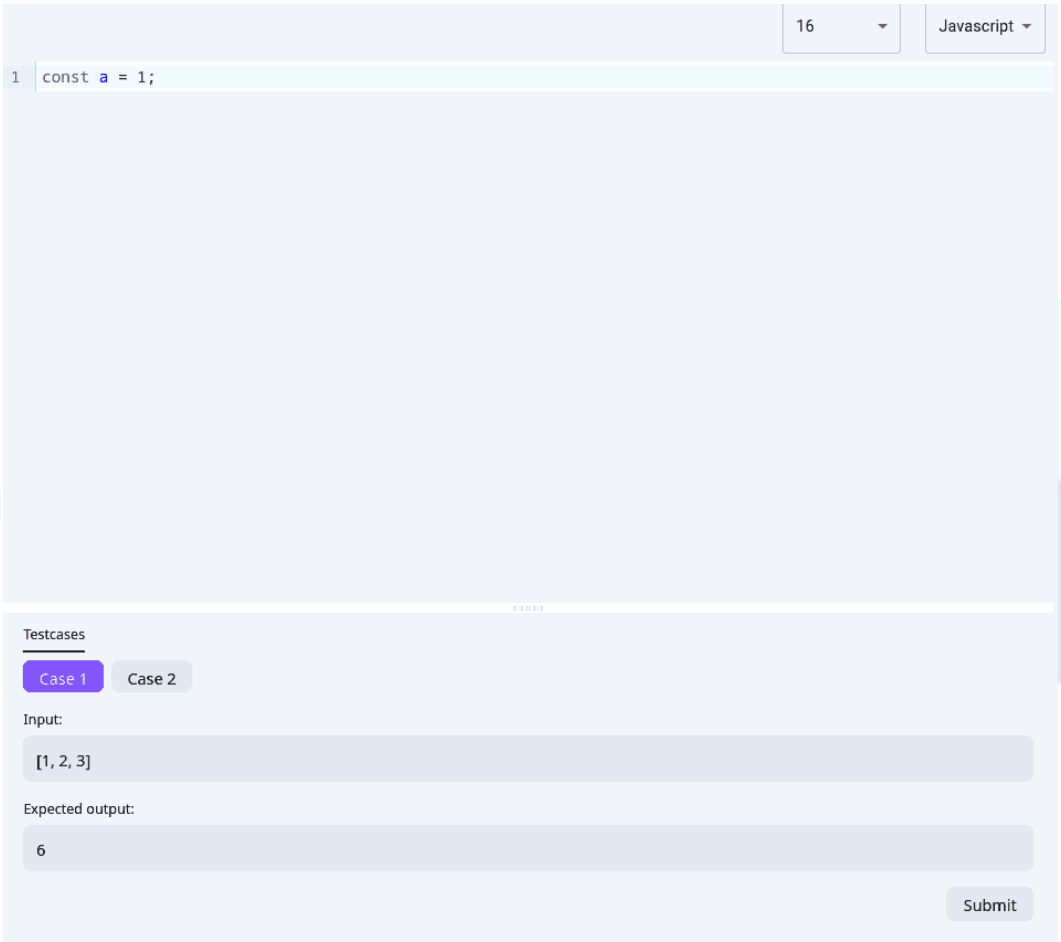
Approach:

A brute force approach involves iterating through the array and checking every pair of elements to see if their sum equals the target. However, this approach would have a time complexity of O(n^2). A more efficient approach involves using a hash table (or dictionary in Python) to store the indices of the elements as we iterate through the array. This way, we can check if the complement of the current element (target - current element) exists in the hash table in constant time, reducing the time complexity to O(n). A brute force approach involves iterating through the array and checking every pair of elements to see if their sum equals the target. However, this approach would have a time complexity of O(n^2). A more efficient approach involves using a hash table (or dictionary in Python) to store the indices of the elements as we iterate through the array. This way, we can check if the complement of the current element (target - current element) exists in the hash table in constant time, reducing the time complexity to O(n)

1.3. Playground Component

- The second half of the screen is split into two halves as well, one for the code editor and one for rendering the test cases and submit the solution for that challenge. Use the same library we used earlier to split the screen, but vertically this time.
- Let's begin with the code editor. For this, we'll be using the powerful tool called react-codemirror. It offers support for a wide range of programming languages, but in our platform, we only support Python and JavaScript. Additionally, it provides support for both light and dark themes!
- At the top, add two configuration options: the programming language and the font size of the code.

- These configuration options will be stored in the Redux store in a slice that we can call workspace.
- You need to implement a DropDown component for these configuration options.
- For the bottom half of the Playground component, we'll display the test cases:
    - The coder can select a test case from different cases for that challenge.
    - We'll show the input text and the expected output text for that test case.
    - A submit button will be used to submit the code to the grading service in the backend later.



2. Update the Routing

- The Workspace page should be accessed via a route **/workspace/:challengeId**, the **challengeId** is a route parameter used to fetch the challenge's data.
- The challenges in the home page link to the workspace page.